Capgemini

Team : S3NTRY HACKERS

EcoRoute AI

AWS Global
Agentic AI Hackathon

In collaboration with aws onova

The Challenge we are facing

Modern logistics teams must juggle cost, carbon reduction, and customer demands amid growing network complexity and new green regulations. Manual planning can't keep up resulting in inefficiencies and compliance risks

Cost Management

Timely Delivery

CO2 emission

Customer Demand

Planning optimal route considering all parameters

Capgemini | aws onova

Who Benefits from EcoRoute AI ?

Primary User Profiles

Logistics Managers & Route Planners

- Instantly simulate and compare eco-friendly, cost-efficient shipment scenarios
- Gain real-time insights into delivery performance and carbon impact
- Make fast, data-driven decisions with full auditability

Sustainability & Compliance Teams

- Track and report carbon emissions across all shipments and routes
- Automate green certification, ESG and GLEC compliance
- Maintain traceable logs for internal and external audits

Carriers, Fleet Operators & 3PLs

- Optimize asset use and load sharing for greater efficiency
- Offer customers certified low-carbon delivery options
- Understand and improve operations through transparent route analytics

Ideal Customers

Enterprises with Sustainability Mandates

E-commerce, FMCG, retail, and manufacturing companies needing to prove ESG compliance and reduce logistics carbon footprint

Large Logistics Providers

3PLs and multimodal carriers managing complex supply networks, under pressure to reduce costs and emissions

Public Sector & Government

City, state, or national agencies running transport and environmental programs, or municipal logistics

Our solution

We have built an ,

Autonomous Green Logistic Optimizer that uses Agentic AI to revolutionize and transport teams plan, compare, and book multimodal shipment for lowest cost and carbon.

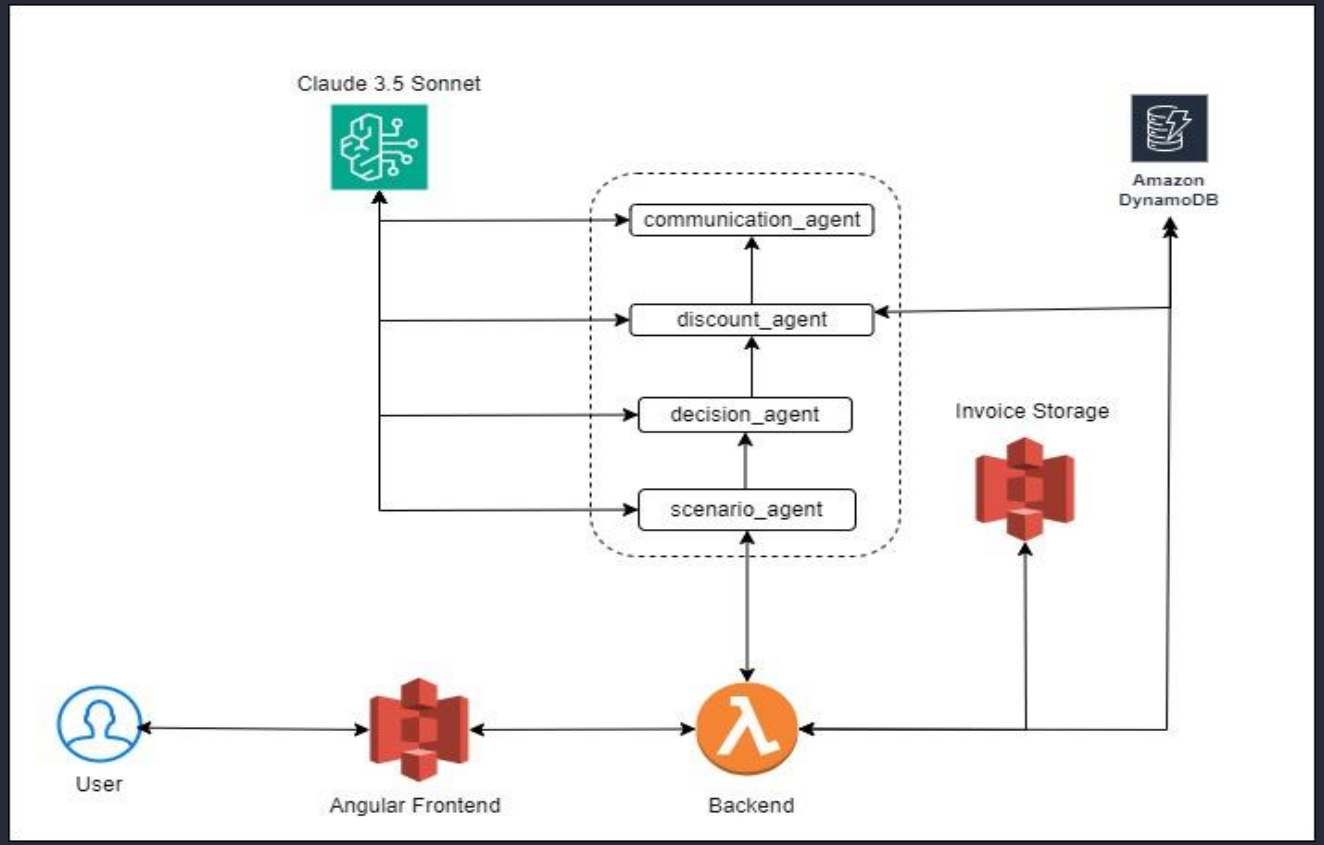
We are enabling -

- Auto-capture shipment, route and preference data
- Simulate and evaluate all possible multimodal scenarios
- Recommending the optimal route and generate compliance ready reports

The built-in smart assistant empowers planners to query shipments route analytics and sustainability scores making logistics not only faster and greener, but also transparent, audible and customer-aligned

Capgemini | aws onova

Solution Architecture



- Use **Angular framework for frontend development.**
- Leverage **Chart.js** for creating interactive data visualizations
- Employ **RxJS** to manage asynchronous data streams effectively

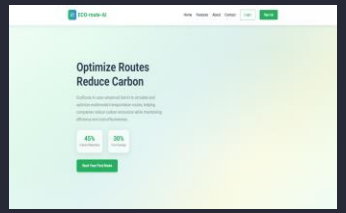
- Utilize **Python FastAPI** for building and managing APIs
- Implement **Strands Agent** to orchestrate agent workflows
- Use **Pydantic** to validate and structure LLM responses

- Integrate **Bedrock (Claude 3.5 sonnet)** for reasoning, decision making .
- Host frontend on S3 and stored shipment invoice
- Maintain user and shipment data in **DynamoDB**

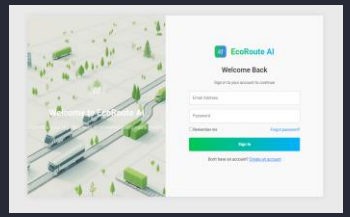
Capgemini | aws onova

How are we doing

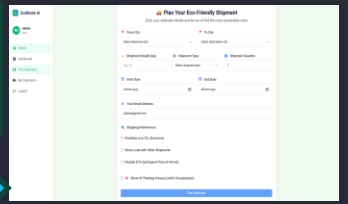
1.Landing Page



2. Login Page



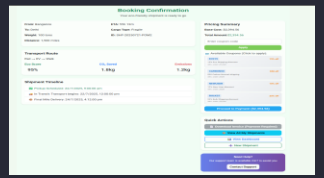
3. Shipment Details



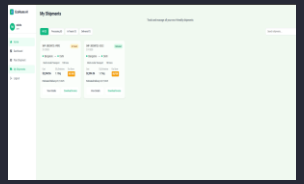
4. AI Processing

5 Route Simulation

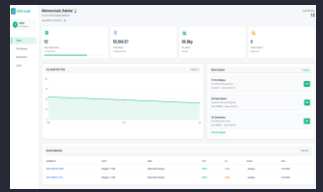
6. Route Confirmation



7. My shipments



8.Dashboard



Timeline & Execution Plan

Laying The Foundation

- Research Market & logistics partner
- Collect CO_2 route & warehouse data
- Align with GLEC & compliance standards
- Design UI(Figma), set up AWS infra (DynamoDB, S3, Bedrock)

Prototyping & Validation

- Build MVP: Agentic simulation, CO_2 calculator
- Develop APIs for scenario suggestions & booking
- Integrate "Green Coupon" logic
- Run internal tests & conduct pilot with 1-2 logistics partners

Productization & Onboarding

- Launch API + dashboard with JWT auth & analytics
- Enable live tracking, reports and scenario comparison
- Deploy chatbot & onboard logistics partners

Scaling & Value Realization

- Add advanced analytics & carbon reporting
- Integrate external APIs(traffic, weather, offsets)
- Launch incentive modules & post-launch support

1 Months	2 Months	1 Months	2 Months	
August 2025	SEPT 2025	NOV 2025(API launch)	DEC 2025	FEB 2026

Revenue Model

Cost-recovery only for first 3-5 pilot customers (build trust, gather feedback) Resource cost: $50K (team, infra), infra Cost:$10K(AWS)	MVP Development & implementation (profit: 10-20%) Resource_infra-400-500k$	Feature Customization & ongoing Support (profit: 15-30%)

Appendix-
Solution Screens

ECO-route-AI

Home Features About Contact Login Sign Up

Optimize Routes Reduce Carbon

EcoRoute AI uses advanced GenAI to simulate and optimize multimodal transportation routes, helping companies reduce carbon emissions while maintaining efficiency and cost-effectiveness.

45%
Carbon Reduction

30%
Cost Savings

Start Your First Route

EcoRoute AI

Create Account

Full Name

Company Name

Email Address

Password

Confirm Password

☐ I agree to the Terms of Service and Privacy Policy

Create Account

Already have an account? Sign in

EcoRoute AI

Welcome Back

Sign in to your account to continue

Email Address

Password

☐ Remember me

Forgot password?

Sign In

Don't have an account? Create an account

🚚 Plan Your Eco-Friendly Shipment

Enter your shipment details and let our AI find the most sustainable route

📍 From City

Select departure city

📍 To City

Select destination city

⚖️ Shipment Weight (kg)

e.g. 10

📦 Shipment Type

Select shipment type

🔢 Shipment Quantity

1

📅 Start Date

dd-mm-yyyy

📅 End Date

dd-mm-yyyy

📧 Your Email Address

raksha@gmail.com

🔍 Shipping Preferences

- [ ] Prioritize Low CO_2 Emissions
- [ ] Share Load with Other Shipments
- [ ] Flexible ETA (Estimated Time of Arrival)
- [ ] 🧠 Show AI Thinking Process (AGUI Visualization)

Plan Shipment

ECO-route AI Processing

Claude 3.5 is optimizing your sustainable routes

100% Complete

Completed

1 **Scenario Agent**
Generating route options

02:18:12 am - 02:18:14 am (2.50s)

Analyzing possible transportation modes for your shipment...

Scenario Agent Analysis

"Completed analysis of 3 route scenarios. Ready to pass to decision agent for optimization."

2 **Decision Agent**
Evaluating and optimizing routes

02:18:17 am - 02:18:19 am (2.50s)

Calculating emissions impact and ranking routes based on your preferences...

Decision Analysis

"Evaluating route options and determining the optimal choice based on preferences"

3 **Discount Agent**
Checking for applicable discounts

02:18:19 am - 02:18:22 am (2.50s)

Searching for eco-discounts that match your routes...

Discount Evaluation

"Analyzing your eco-impact score and discount eligibility..."

4 **Communication Agent**
Finalizing recommendation

02:18:22 am - 02:18:24 am (2.50s)

Preparing final route recommendations with Claude 3.5...

Final Recommendation

"Generating a human-readable summary of the selected route"

3 **Discount Agent**
Checking for applicable discounts

02:18:19 am - 02:18:22 am (2.50s)

Searching for eco-discounts that match your routes...

Discount Evaluation

"Analyzing your eco-impact score and discount eligibility..."

4 **Communication Agent**
Finalizing recommendation

02:18:22 am - 02:18:24 am (2.50s)

Preparing final route recommendations with Claude 3.5...

Final Recommendation

"Generating a human-readable summary of the selected route"

Powered by Claude 3.5 Sonnet & ECO-route AI

Processing complete!

View Route Options

Route Simulation Results

AI-optimized multimodal transportation scenarios for your shipment

Route: **Bangalore → Mumbai** | Weight: **1000 tons** | Type: **Chemicals**

AI Route Recommendation

This multi-modal transport option capitalizes on rail's efficiency over long distances combined with flexible last-mile delivery on foot, minimizing carbon emissions while maintaining strong value and adaptability for certain customer needs. By seamlessly linking nationwide infrastructure, specialized deliveries to remote or last-minute destinations can be reliably accommodated through dynamic interchanging of transportation modes, providing a resilient solution for time sensitive shipments or changing circumstances.

Sort & Compare Routes

Sort by: Recommended | Eco Score | $ Cost | Time

AI Recommended - Optimal Balance

Transport Mode

Rail → Highway → Walk

Estimated Time
42h 42m

Total Cost
$873.18

CO_2 Emissions **0g**

Excellent Impact

CO_2 Impact

0g CO_2

Eco Score
95
ECO

Reliability: 95%

This multi-modal transport route leverages the efficiency of rail over long distances, flexibility of highway routes near destinations, and environmental sustainability of pedestrian mobility in dense urban areas. By seamlessly integrating three modes of low-carbon travel, it provides a cost-effective option with minimal risk of delays from traffic or accidents. For certain schedules with time buffers or destinations near transit hubs, this route's multi-pronged connectivity and zero emissions present a compelling alternative to direct motorized options, strengthening competitive positioning in environmentally-conscious and urban markets.

Choosing the rail-highway-walk route from Bangalore to Mumbai reduces carbon emissions by 100%, having no CO2 impact on the environment and helping minimize pollution.

Transport Mode

Rail → Highway

Estimated Time

43h 54m

Total Cost

$7,981.88

CO_2 Emissions **258g**

High Impact

CO_2 Impact

258g CO_2

Eco Score

48

ECO

Reliability: 70%

This hybrid rail-highway route from Bangalore to Delhi offers significant flexibility and risk mitigation advantages. By combining the speed and reliability of rail transport over long distances with the last-mile connectivity of highways, this option provides a robust multi-modal solution well-suited for time-sensitive or high-value cargo needing alternative transport options in the event of disruptions to primary routes. The ability to switch between rail and road transport mid-route enhances operational resilience for just-in-time deliveries or those transporting perishable goods highly dependent on predictable arrival times.

View Details

Select Route

Transport Mode

Highway

Estimated Time

43h 30m

Total Cost

$13,569.19

CO_2 Emissions **387g**

High Impact

CO_2 Impact

387g CO_2

Eco Score

45

ECO

Reliability: 70%

This longer highway route between Bangalore and Delhi offers significant flexibility and resilience advantages. At under half the cost and carbon of air transport, it allows for on-route adjustments, backup plans in the case of delays, and fulfills mission-critical deliveries even during periods of airspace restrictions or airport closures. For time-insensitive yet mission-essential cargo during uncertain conditions, or projects with fluctuating schedules, this option maintains business continuity while spending transportation dollars far more efficiently than other modes.

View Details

Select Route

Booking Confirmation

Your eco-friendly shipment is ready to go

From: Bangalore

To: Delhi

Weight: 1000 tons

Distance: 1,500 miles

ETA: 50h 24m

Cargo Type: Food Products

ID: SHP-20250721-66PW

Transport Route

Rail → Highway → Walk

Eco Score	CO_2 Saved	Emissions
95%	1.8kg	1.2kg

Shipment Timeline

- ✅ Pickup Scheduled: 22/7/2025, 9:00:00 am
- 🚚 In Transit: Transport begins: 22/7/2025, 12:00:00 pm
- 📦 Final Mile Delivery: 24/7/2025, 11:24:00 am

Pricing Summary

Base Cost: $2,203.00

Total Amount:$2,203.00

Enter coupon code

Apply

💳 Available Coupons (Click to apply):

EC015 — 15% off
15% Eco Shipping discount
Min. order: $1,000

CARBON50 — $50 off
$50 Carbon Neutral shipping
Min. order: $500

NEWUSER — 10% off
10% New User discount
Min. order: $500

BULK25 — 25% off
25% Bulk Shipping discount
Min. order: $3,000

Proceed to Payment ($2,203.00)

Quick Actions

- 📄 Download Invoice (Payment Required)
- 📦 View All My Shipments
- 📊 View Dashboard
- ➕ New Shipment

Need Help?

Our support team is available 24/7 to assist you

Contact Support

EcoRoute AI

RA raksha
user

- Home
- Dashboard
- Plan Shipment
- My Shipments
- Logout

My Shipments

Track and manage all your eco-friendly shipments

All (2) | Processing (1) | In Transit (0) | Delivered (1)

SHP-20250721-66PW — Processing
21/7/2025

Bangalore → Delhi

Multi-modal Transport 1000 tons

Cost	CO_2 Emissions	Eco Score
$2,203.00	1.15kg	85/100

Estimated Delivery:23/7/2025

View Details | Download Invoice

SHP-20250721-1875 — Delivered
21/7/2025

Bangalore → Delhi

Multi-modal Transport 1000 tons

Cost	CO_2 Emissions	Eco Score
$2,203.00	1.15kg	85/100

Estimated Delivery:23/7/2025

View Details | Download Invoice

ECO-route

raksha
Eco Champion

- Home
- Plan Shipment
- My Shipments
- Logout

Welcome back, Raksha!

Your eco-friendly shipping dashboard

Last updated: 2:13:38 am

Total Shipments
12

92
Green Impact Score
+12% this month

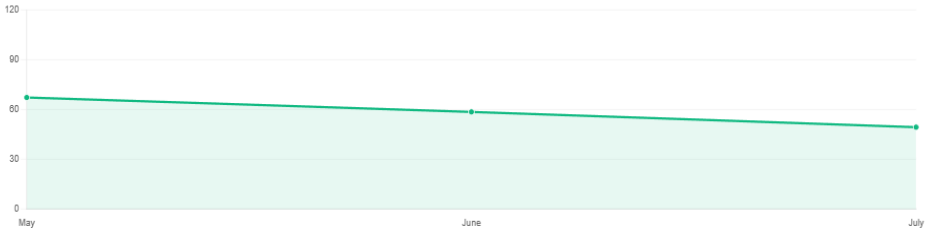
$5,003.4
Total Savings
vs traditional routes

86.8kg
CO_2 Saved
This year

8
Active Coupons
Ready to use

CO_2 Saved Over Time

3 Months



Green Coupons

3 Available

15% Eco Shipping
For routes with 90%+ green score
Code: ECO15 Expires: 2024-02-28
15%

$50 Carbon Neutral
Complete 5 carbon-neutral shipments
Code: CARBON50 Expires: 2024-03-15
$50

20% Green Routes
Eco-friendly transport modes
Code: GREEN20 Expires: 2024-03-31
20%

See All Coupons

Recent Shipments

View All

SHIPMENT ID	ROUTE	MODE	COST	CO_2	STATUS	DATE
SHP-20250721-66PW	Bangalore – Delhi	Multi-modal Transport	$2203	0.0kg	Confirmed	07-20-2025
SHP-20250721-1875	Bangalore – Delhi	Multi-modal Transport	$2203	0.0kg	Confirmed	07-20-2025